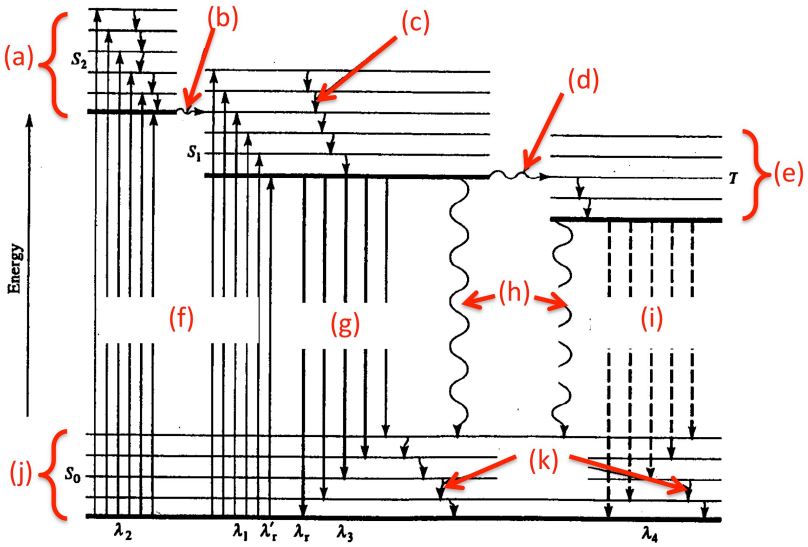
Analytical Chemistry Cumulative Exam
February 7, 2020

Fluorescence Spectroscopy (in the UV through visible wavelength range): Theory, Instrumentation, and Applications

1. (10 points) Name the following processes (b, c, d, f, g, h, i, k) that correspond to energy transitions for electrons in a photoluminescent molecule. Some of the letters (a, j, e) refer to electron states. Please name those, too. (Note that (c) and (k) represent the same kind of transition.)



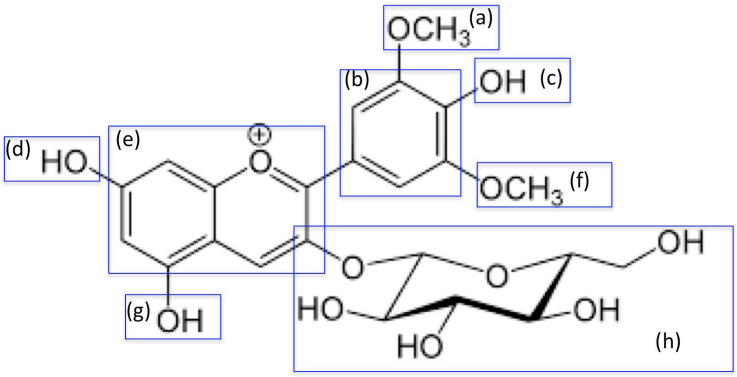
2. (5 points) Which of the transitions can take as long as 10^2 s to complete? Why?

3. (10 points) Define quantum yield. Write an equation expressing quantum yield as a function of the rate constants for the processes shown in the figure above. You can include other processes if you think that some have been left out in the diagram.

4. (27 points) Below are several properties or situations involving a fluorescing molecule. Indicate what changes of these properties (an "increase" or a "decrease") could cause an ***improvement*** in the probability of the fluorescence of the molecule. If a change in the property is not expected to increase or decrease the fluorescence, then write "a change does not affect fluorescence". Explain the reasons for your answers.

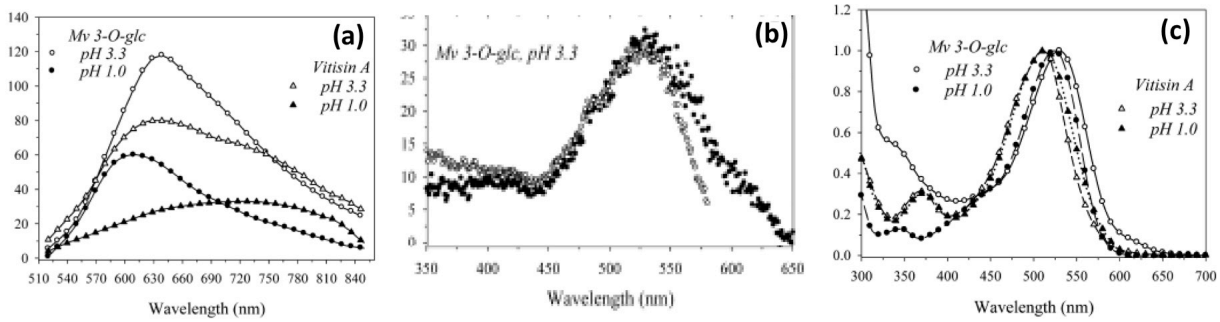
(a) energy of the π* to π transition
(b) molar absorptivity
(c) rigidity
(d) temperature
(e) viscosity of solvent
(f) presence of paramagnetic species in surrounding solution
(g) the number of resonant forms
(h) modification with halogen substituents
(i) modification with carboxylic acid substituents

5. (10 points) This structure is of one of the colorful pigments called anthocyanins that are found in red wine. It is **malvidin-3-O-glucoside**. This molecule is usually quantified by UV-Vis, and there are studies where it has been analyzed with FTIR. However, this molecule should also fluoresce. It turns out that anthocyanins are actually only weakly fluorescent in solution, and thus, fluorescence spectroscopy of red wine has not been extensively studied. However, *Agati et al.** investigated this possibility and found that the older wines have aggregates of anthocyanines, which more strongly fluoresce and can be the basis of a method to follow wine aging.



Which blue-boxed and labeled portion of the molecule is mainly responsible for its ability to fluoresce? Explain your choice. Why is it a weak fluorophore?

6. (11 points) Below in random order are an absorbance spectrum, an excitation spectrum, and a fluorescence spectrum of malvidin-3-O-glucoside (abbreviated Mv 3-O-glc) at two different pH values from *Agati et al.* **Assign each spectrum and explain why you chose it** . (Note that vitisin A is a pyruvic acid adduct of Mv 3-O-glc and its spectra should be ignored in answering this question.)



** Giovanni Agati, Paolo Matteini, Joana Oliveira, Victor de Freitas, and Nuno Mateus, "Fluorescence Approach for Measuring Anthocyanins and Derived Pigments in Red Wine", J. Agric. Food Chem. 2013, 61,42, 10156-10162*

7. (15 Points) Draw a simple block diagram for the instrumental components of a traditional fluorescence spectrometer. Label each block and show (with arrow(s)) the path(s) the radiation takes. (Include the relative location of the following, if relevant: light source(s), wavelength selector(s), sample, detector(s), computer readout and acquisition.)

8. (12 points) Advanced fluorescence techniques. Explain how the following two advanced fluorescence techniques work and the kinds of information that they can provide about a molecule, chemical system, or chemical process.
(a) Fluorescence correlation spectroscopy (FCS)
(b) Fluorescence resonance energy transfer (FRET)